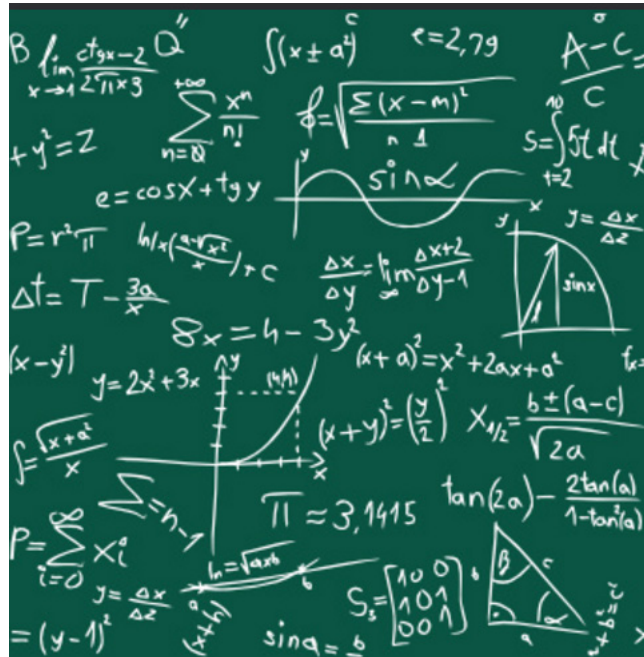


Fundamental limits and asymptotic estimates



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- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Fundamental limits

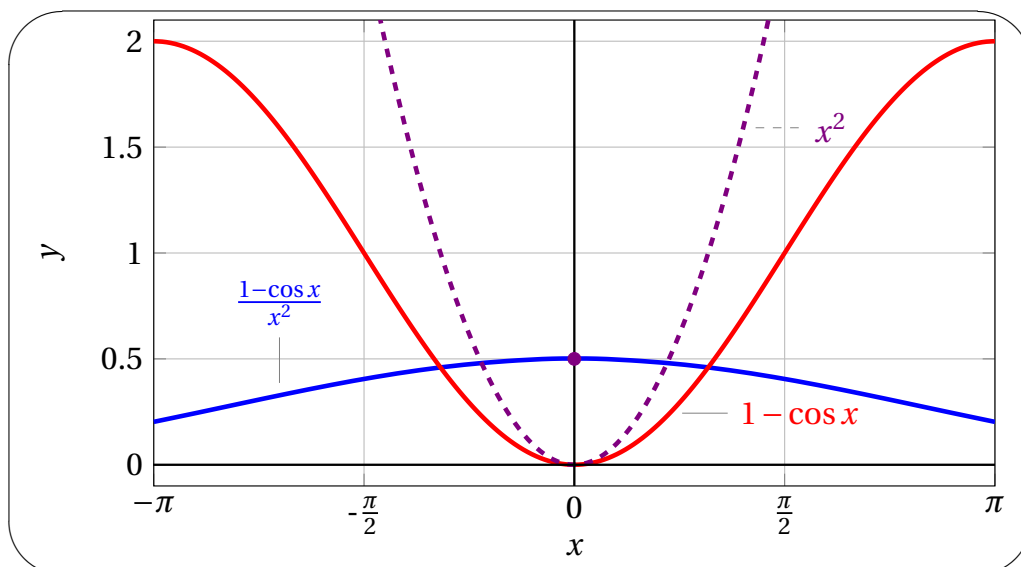
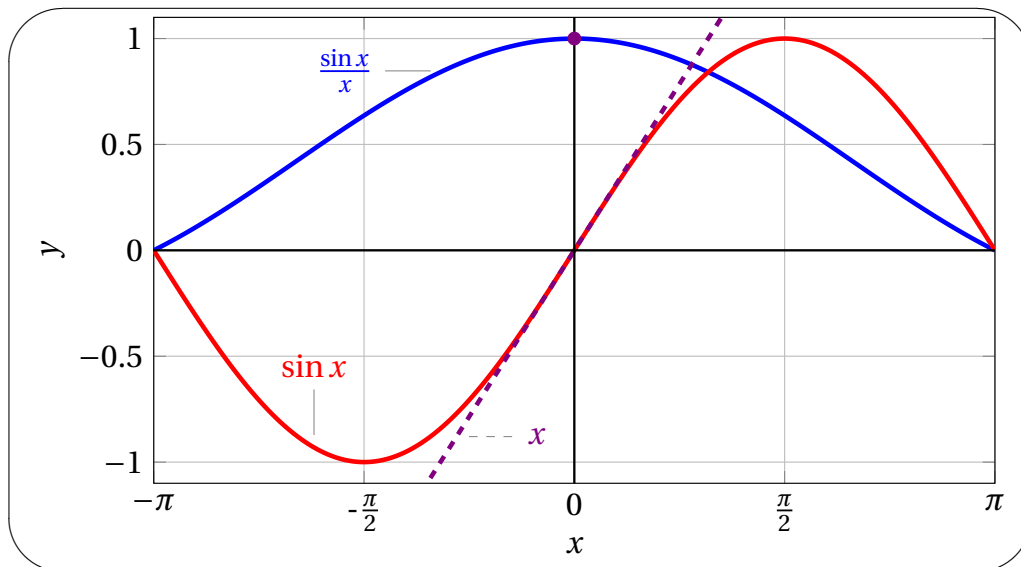
- We now look at some common techniques for computing limits that combine the general theorems on limits with the use of some fundamental limits of certain elementary functions.

1.1 Fundamental limits of sine and cosine

- We want to compute

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \left[\frac{0}{0} \right] \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \left[\frac{0}{0} \right]$$

which are indeterminate forms.



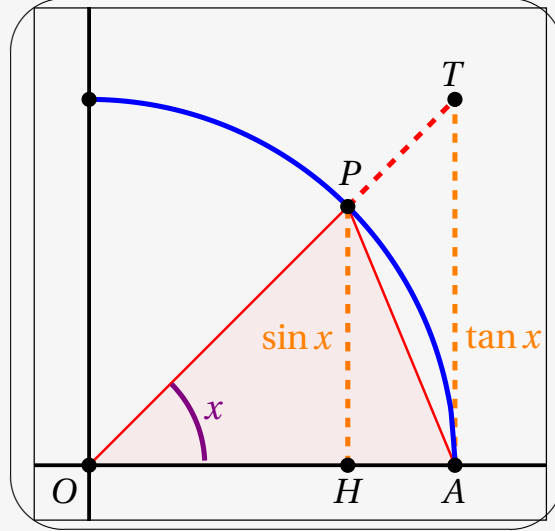
Lemma 1

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad (1)$$

Proof. The functions $\sin x$ and x are odd functions, hence $\frac{\sin x}{x}$ is an even function. Therefore it is sufficient to compute

$$\lim_{x \rightarrow 0^+} \frac{\sin x}{x}$$

We have:



$$HP = \sin x \quad AT = \tan x \quad \widehat{AP} = x$$

The area of the triangle OPA is smaller than that of the circular sector OPA , which in turn is smaller than that of the triangle OTA . It follows that

$$\underbrace{\frac{1}{2} \cdot 1 \cdot \sin x}_{\text{area triangle } OPA} \leq \underbrace{\frac{1}{2} \cdot 1 \cdot x}_{\text{area circ. sector } OPA} \leq \underbrace{\frac{1}{2} \cdot 1 \cdot \tan x}_{\text{area triangle } OTA}$$

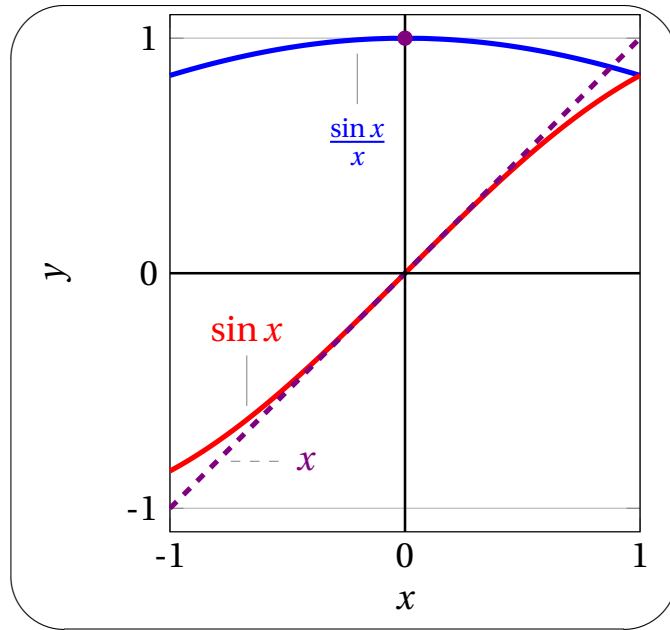
that is, for $x \in (0, \frac{\pi}{2})$:

$$\sin x < x < \tan x$$

Dividing by $\sin x$, which is positive because $x \in (0, \frac{\pi}{2})$, we have

$$1 < \frac{x}{\sin x} < \frac{1}{\cos x}, \quad \forall x \in (0, \frac{\pi}{2}) \quad \text{that is} \quad \cos x < \frac{\sin x}{x} < 1, \quad \forall x \in (0, \frac{\pi}{2})$$

By the comparison theorem, since $\lim_{x \rightarrow 0} \cos x = 1$, the lemma follows. □



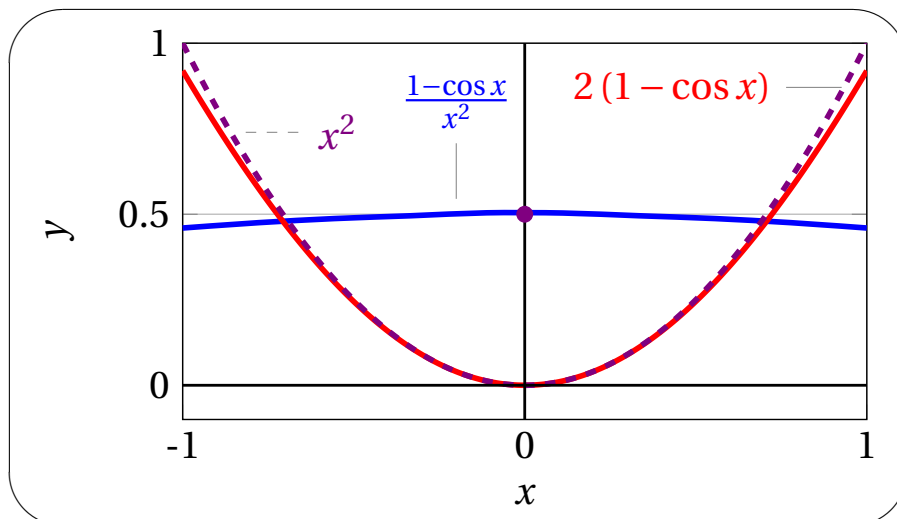
Lemma 2

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2} \quad (2)$$

Proof. We consider

$$\frac{1 - \cos x}{x^2} = \frac{1 - \cos^2 x}{x^2 (1 + \cos x)} = \left(\frac{\sin x}{x} \right)^2 \frac{1}{1 + \cos x}$$

hence, since $\frac{\sin x}{x} \rightarrow 1$ and $(1 + \cos x) \rightarrow 2$ as $x \rightarrow 0$, the claim follows. $\square$



1.2 Continuous extension of a function

- Based on the limit (1) just proved, the functions $f(x) = \frac{\sin x}{x}$ and $g(x) = \frac{1-\cos x}{x^2}$, initially not defined at $x = 0$, can be extended by continuity also at $x = 0$, by setting

$$f(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases} \quad g(x) = \begin{cases} \frac{1-\cos x}{x^2} & \text{if } x \neq 0 \\ \frac{1}{2} & \text{if } x = 0 \end{cases}$$

The functions f and g defined in this way are continuous also at $x = 0$.

- if a function $f(x)$ is not defined at x_0 but the finite limit exists

$$\lim_{x \rightarrow x_0} f(x) = \ell$$

the function can be extended by continuity also at x_0 , by defining

$$f(x_0) = \ell$$

- If instead the function f has at x_0 a jump discontinuity, a vertical asymptote, or in any case does not have a finite limit, it is not possible to make it continuous at x_0 by changing its definition at a single point.

1.3 Other fundamental limits

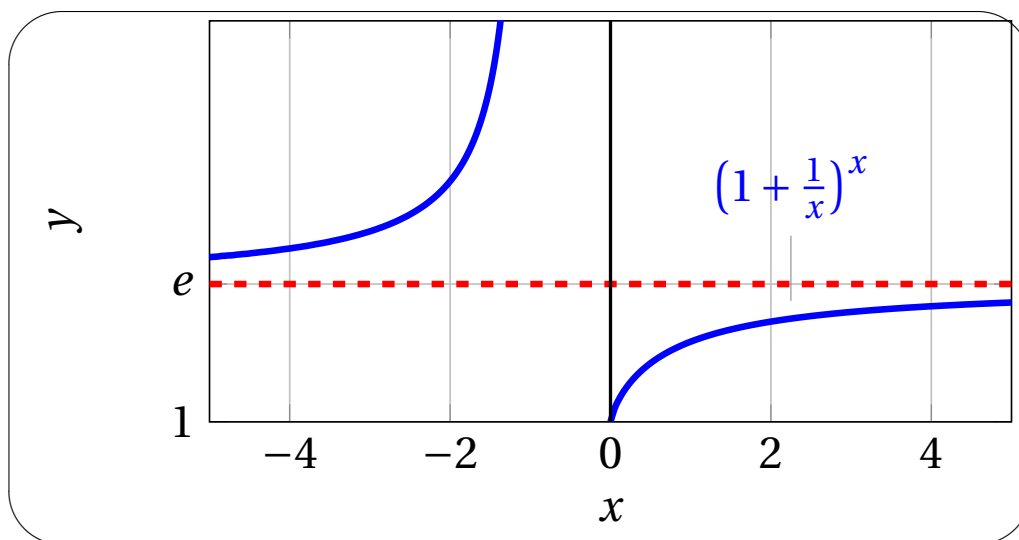
- We know that for every sequence $\{a_n\}$ diverging to $+\infty$ or $-\infty$ we have

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{a_n}\right)^{a_n} = e$$

- By the sequential definition of the limit of a function, this fact immediately implies the next fundamental limit

Lemma 3

$$\lim_{x \rightarrow \pm\infty} \left(1 + \frac{1}{x}\right)^x = e \quad (3)$$



If $\eta(x)$ is a function that tends to infinity^a (i.e., it is an infinity: $\eta(x) \rightarrow \pm\infty$), we have:

$$\left(1 + \frac{1}{\eta(x)}\right)^{\eta(x)} \rightarrow e \quad (4)$$

^ait does not matter what x tends to

Example 1: Limits of functions tending to e

1.

$$\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{2x^2 - 10}\right)^{2x^2 - 10} = e$$

since with $\eta(x) = 2x^2 - 10$ we have

$$\eta(x) \rightarrow +\infty \text{ as } x \rightarrow +\infty.$$

2.

$$\lim_{x \rightarrow 0^+} \left(1 + \frac{1}{1/x}\right)^{1/x} = e$$

since with $\eta(x) = \frac{1}{x}$ we have

$$\eta(x) \rightarrow +\infty \text{ as } x \rightarrow 0^+.$$

From the fundamental limit (3) three more can be deduced

Corollary 1

$$\lim_{y \rightarrow 0} \frac{\log(1+y)}{y} = 1 \quad (5)$$

Proof. Taking logarithms in (3), we obtain

$$\log\left(1 + \frac{1}{x}\right)^x = x \log\left(1 + \frac{1}{x}\right)$$

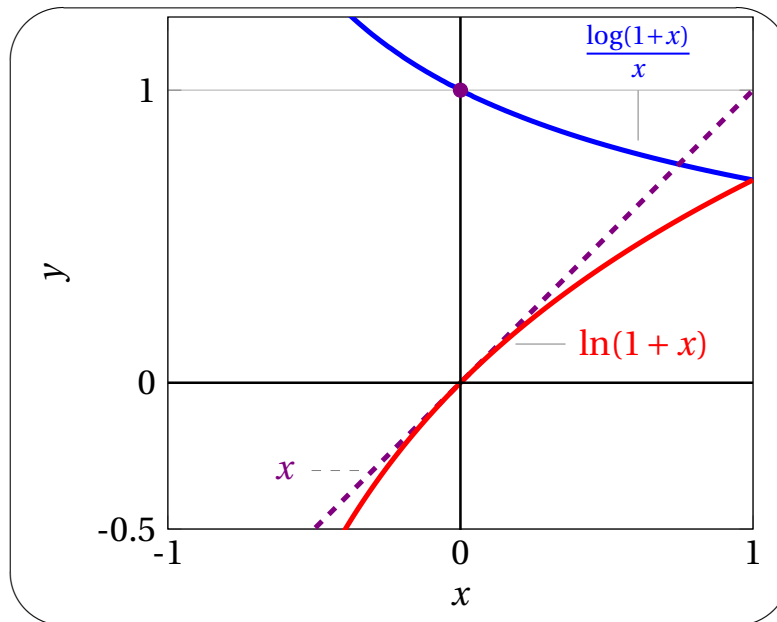
hence

$$\lim_{x \rightarrow \pm\infty} x \log\left(1 + \frac{1}{x}\right) = \log e = 1.$$

Now, if we set $y = \frac{1}{x}$, $x \rightarrow \pm\infty$ is equivalent to $y \rightarrow 0^\pm$, and the last limit can therefore be rewritten in the following form:

$$\frac{\log(1+y)}{y} \rightarrow 1 \text{ as } y \rightarrow 0.$$

□



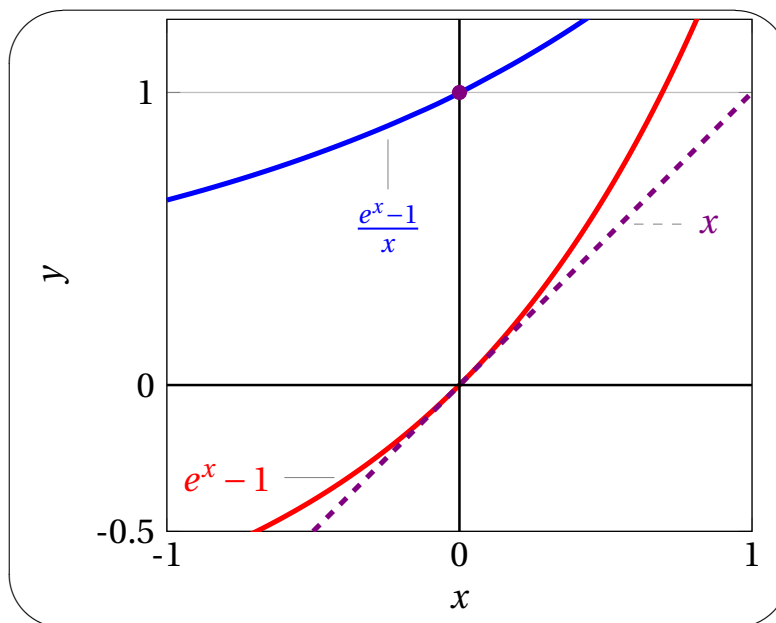
Corollary 2

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \quad (6)$$

Proof. If in (5) we instead set $y = e^x - 1$, $y \rightarrow 0$ is equivalent to $x \rightarrow 0$, and substituting we obtain

$$\frac{\log e^x}{e^x - 1} = \frac{x}{e^x - 1} \rightarrow 1 \text{ as } x \rightarrow 0.$$

Taking reciprocals we obtain the fundamental limit. □



Example 2: Fundamental limit

We compute:

$$\lim_{x \rightarrow 0} \frac{e^{-x} - 1}{x}$$

We set $z = -x$; $x \rightarrow 0$ is equivalent to $z \rightarrow 0$, and substituting we obtain:

$$\lim_{x \rightarrow 0} \frac{e^{-x} - 1}{x} = \lim_{z \rightarrow 0} \frac{e^z - 1}{-z} = - \lim_{z \rightarrow 0} \underbrace{\frac{e^z - 1}{z}}_{\rightarrow 1} = -1$$

Corollary 3

$$\lim_{x \rightarrow 0} \frac{(1+x)^\alpha - 1}{x} = \alpha \quad \text{with } \alpha \in \mathbb{R} \quad (7)$$

Proof. If in (5) we instead set $y = (1+x)^\alpha - 1$, with α any real exponent, then $x \rightarrow 0$ is equivalent to $y \rightarrow 0$ and we have:

$$\begin{aligned} \frac{\log(1+y)}{y} &= \frac{\log[1 + (1+x)^\alpha - 1]}{(1+x)^\alpha - 1} = \frac{\alpha \log(1+x)}{(1+x)^\alpha - 1} \\ &= \frac{\alpha x}{(1+x)^\alpha - 1} \cdot \frac{\log(1+x)}{x} \rightarrow 1 \text{ as } x \rightarrow 0. \end{aligned}$$

But since also

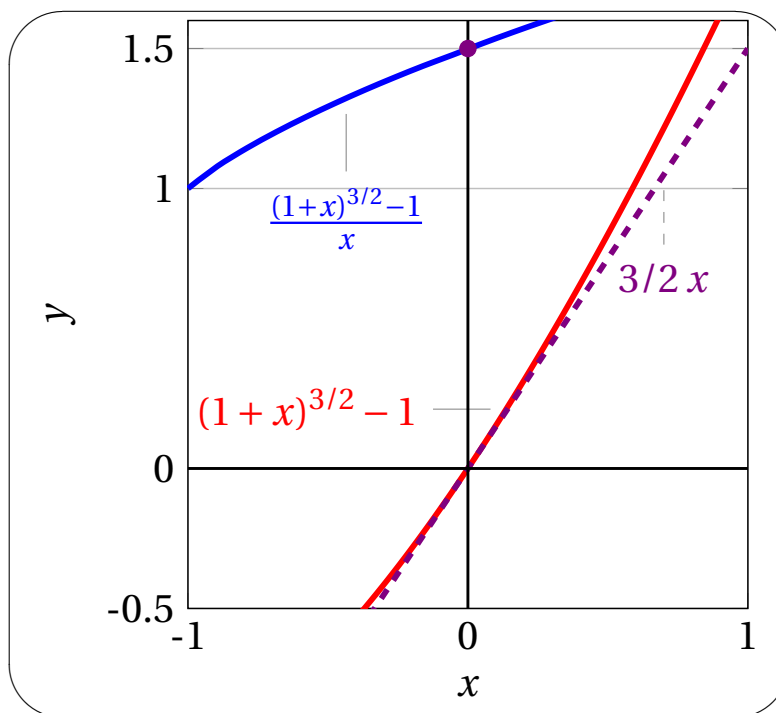
$$\frac{\log(1+x)}{x} \rightarrow 1 \text{ as } x \rightarrow 0,$$

then

$$\frac{\alpha x}{(1+x)^\alpha - 1} \rightarrow 1 \text{ as } x \rightarrow 0.$$

Taking reciprocals and then multiplying by α we obtain the fundamental limit.

□



2 Asymptotic estimates

Definition 1: Asymptotic functions

Two functions f, g are said to be **asymptotic** as $x \rightarrow c$ if

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = 1$$

and we write $f \sim g$ as $x \rightarrow c$.

- The asymptotic symbol for functions enjoys all the properties stated for sequences

As $x \rightarrow 0$ we have (from the fundamental limits):

$$\sin x \sim x \quad (8a)$$

$$\log(1+x) \sim x \quad (8b)$$

$$\cos x \sim 1 - \frac{1}{2}x^2 \quad (8c)$$

$$e^x \sim 1+x \quad (8d)$$

$$(1+x)^\alpha \sim 1+\alpha x \quad \text{with } \alpha \in \mathbb{R} \quad (8e)$$

- The functions $\sim x$ behave, to a first approximation or to first order, like x as $x \rightarrow 0$.

If $\varepsilon(x)$ is a function that tends to zero^a (i.e., it is an infinitesimal: $\varepsilon(x) \rightarrow 0$), we have:

$$\sin(\varepsilon(x)) \sim \varepsilon(x) \quad (9a)$$

$$\log(1+\varepsilon(x)) \sim \varepsilon(x) \quad (9b)$$

$$\cos(\varepsilon(x)) \sim 1 - \frac{1}{2}\varepsilon^2(x) \quad (9c)$$

$$e^{\varepsilon(x)} \sim 1+\varepsilon(x) \quad (9d)$$

$$(1+\varepsilon(x))^\alpha \sim 1+\alpha \varepsilon(x) \quad \text{with } \alpha \in \mathbb{R} \quad (9e)$$

^ait does not matter what x tends to

- Formulas (9) follow from formulas (8) simply by a change of variable

$$x = \varepsilon(x)$$

Example 3: Limits with asymptotic functions

$$\lim_{x \rightarrow 1} \frac{(x-1)^2}{e^{3(x-1)^2} - 1} = \left[\frac{0}{0} \right]$$

We use the estimate

$$e^{\varepsilon(x)} - 1 \sim \varepsilon(x)$$

with

$$\varepsilon(x) = 3(x-1)^2 \text{ and } \varepsilon(x) \rightarrow 0 \text{ as } x \rightarrow 1$$

Then, as $x \rightarrow 1$, we have

$$e^{3(x-1)^2} - 1 \sim 3(x-1)^2.$$

and

$$\lim_{x \rightarrow 1} \frac{(x-1)^2}{e^{3(x-1)^2} - 1} = \lim_{x \rightarrow 1} \frac{(x-1)^2}{3(x-1)^2} = \frac{1}{3}$$

Example 4: Limits with asymptotic functions

$$\lim_{x \rightarrow 0} \frac{\log(1+2x)}{\sin 3x} = \left[\frac{0}{0} \right]$$

We use the estimate

$$\log(1+\varepsilon(x)) \sim \varepsilon(x)$$

with

$$\varepsilon(x) = 2x \text{ and } \varepsilon(x) \rightarrow 0 \text{ as } x \rightarrow 0.$$

Hence, as $x \rightarrow 0$ we have

$$\log(1+2x) \sim 2x$$

Now we also use the estimate

$$\sin \varepsilon(x) \sim \varepsilon(x)$$

with

$$\varepsilon(x) = 3x \text{ and } \varepsilon(x) \rightarrow 0 \text{ as } x \rightarrow 0$$

Hence, as $x \rightarrow 0$ we have

$$\sin 3x \sim 3x$$

Then, as $x \rightarrow 0$, we have

$$\frac{\log(1+2x)}{\sin 3x} \sim \frac{2x}{3x}$$

and

$$\lim_{x \rightarrow 0} \frac{\log(1+2x)}{\sin 3x} = \lim_{x \rightarrow 0} \frac{2x}{3x} = \frac{2}{3}$$

Example 5: Limits with asymptotic functions

$$\lim_{x \rightarrow +\infty} \sqrt[3]{x^3 + 2x^2 + 1} - x = [+ \infty - \infty]$$

We rewrite,

$$\sqrt[3]{x^3 + 2x^2 + 1} - x = x \left(\sqrt[3]{1 + \left(\frac{2}{x} + \frac{1}{x^3} \right)} - 1 \right)$$

We use the estimate

$$\sqrt[3]{1 + \varepsilon(x)} - 1 \sim \frac{1}{3} \varepsilon(x)$$

with

$$\varepsilon(x) = \left(\frac{2}{x} + \frac{1}{x^3} \right) \text{ and } \varepsilon(x) \rightarrow 0 \text{ as } x \rightarrow +\infty.$$

Then, as $x \rightarrow +\infty$, we have

$$\sqrt[3]{x^3 + 2x^2 + 1} - x \sim x \left[\frac{1}{3} \left(\frac{2}{x} + \frac{1}{x^3} \right) \right]$$

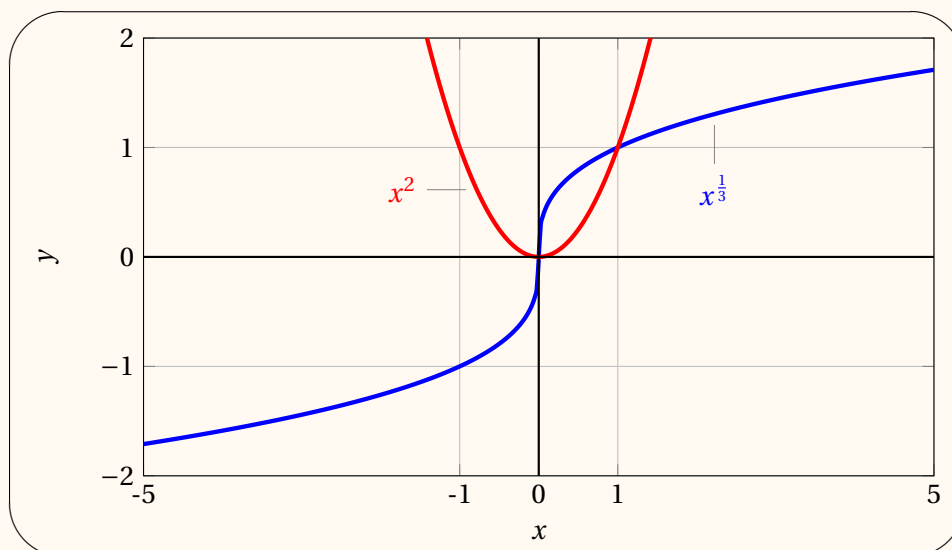
and

$$\lim_{x \rightarrow +\infty} \sqrt[3]{x^3 + 2x^2 + 1} - x = \lim_{x \rightarrow +\infty} x \left[\frac{1}{3} \left(\frac{2}{x} + \frac{1}{x^3} \right) \right] = \frac{2}{3}$$

3 Asymptotic estimates and graphs

- Asymptotic estimates are useful not only for computing limits, but also for sketching the qualitative graph of a function in a neighborhood of a given point, or as $x \rightarrow \pm\infty$.

Example 6: Known graphs



Example 7: Qualitative graph

We now want to study the qualitative graph of the function (the sum of the previous two):

$$f(x) = \sqrt[3]{x} + x^2 = x^{\frac{1}{3}} + x^2$$

- the function is defined and continuous on all of $\mathbb{R}$;
- as $x \rightarrow \pm\infty$, $f(x) \sim x^2$, since

$$\lim_{x \rightarrow \pm\infty} \frac{x^{\frac{1}{3}} + x^2}{x^2} = 1$$

Therefore $f(x) \rightarrow +\infty$ as $x \rightarrow \pm\infty$; moreover, for x large in absolute value, its graph will be similar to that of x^2 .

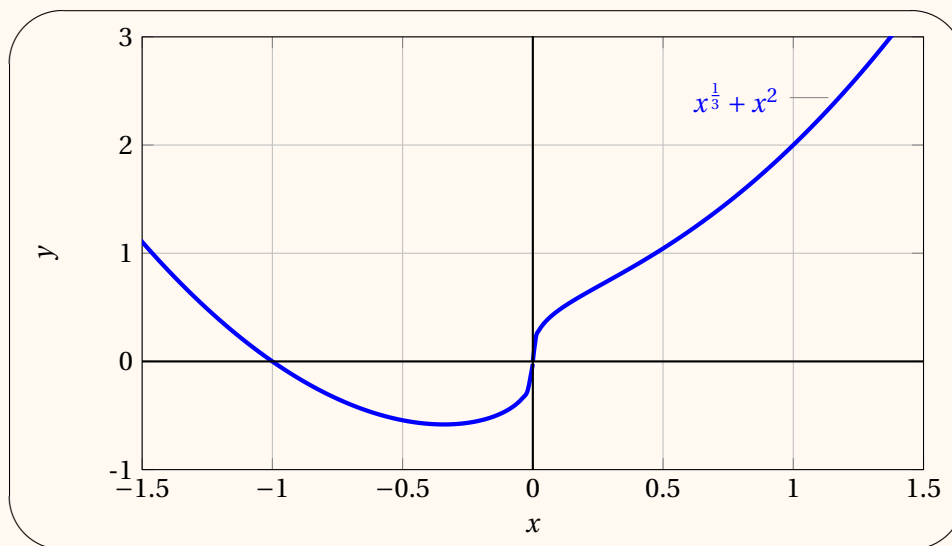
- Moreover, the function vanishes at $x = 0$ and, as $x \rightarrow 0$, $f(x) \sim x^{\frac{1}{3}}$, since

$$\lim_{x \rightarrow 0} \frac{x^{\frac{1}{3}} + x^2}{x^{\frac{1}{3}}} = 1$$

therefore, in a neighborhood of $x = 0$, its graph will be similar to that of $x^{\frac{1}{3}}$; in particular, it will have a vertical tangent at the origin.

Example 8: Actual graph

$$f(x) = \sqrt[3]{x} + x^2 = x^{\frac{1}{3}} + x^2$$



- Often the behavior of a function in a neighborhood of a point (for example, the fact that it has a vertical or horizontal tangent) can be predicted from a suitable asymptotic estimate.
- The asymptotic estimate allows us to sketch the qualitative graph of f (in a neighborhood of the point) by comparison with that of a known function (for example, a power with

rational exponent).

- Analogous estimates are useful as $x \rightarrow \pm\infty$.

Growth of a function at infinity

- Suppose we want to sketch the graph of a function f that, as $x \rightarrow +\infty$ (or $-\infty$), tends to $+\infty$ (or $-\infty$).
- To describe the speed at which the function tends to infinity, the following notions are useful: we say that, as $x \rightarrow +\infty$,

$$f \text{ has } \begin{cases} \text{superlinear growth} \\ \text{linear growth} \\ \text{sublinear growth} \end{cases} \quad \text{if} \quad \lim_{x \rightarrow +\infty} \frac{f(x)}{x} = \begin{cases} \pm\infty \\ m \text{ (finite and different from 0)} \\ 0 \end{cases}$$

- Analogous definitions are given for $x \rightarrow -\infty$.
- Only when a function has linear growth is it possible for it to have an oblique asymptote

Example 9: Growth of a function at infinity

For example, as $x \rightarrow +\infty$

- Superlinear growth: exponentials a^x and powers x^a with $a > 1$.
- Sublinear growth: logarithms $\log_a x$ and powers x^a with $0 < a < 1$.

Example 10: Growth of a function at infinity

The function

$$f(x) = 2x + e^x + e^{\frac{1}{x}}$$

- as $x \rightarrow +\infty$ it is asymptotic to e^x ; therefore it tends to $+\infty$ with superlinear growth;
- as $x \rightarrow -\infty$ it is asymptotic to $2x$; therefore it tends to $-\infty$ linearly
- since

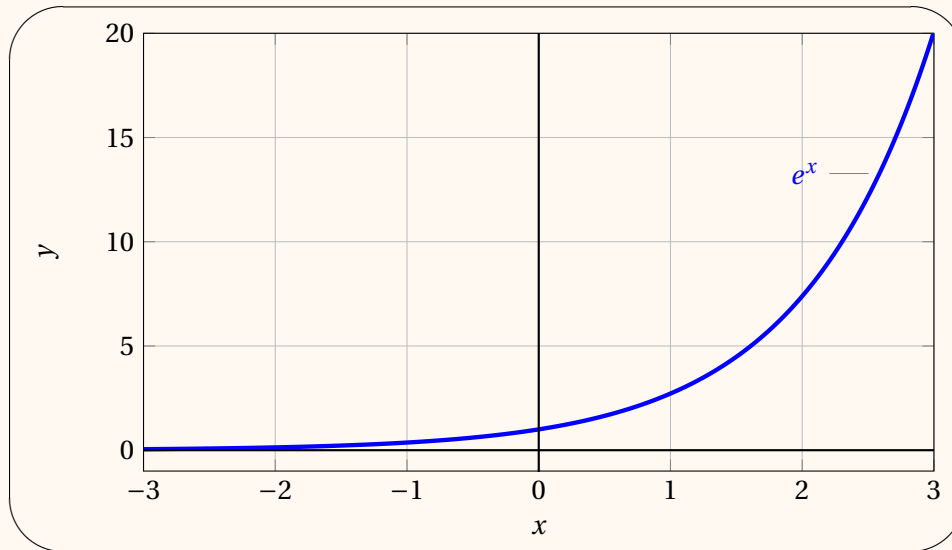
$$\lim_{x \rightarrow -\infty} [f(x) - 2x] = 1$$

the function has the oblique asymptote

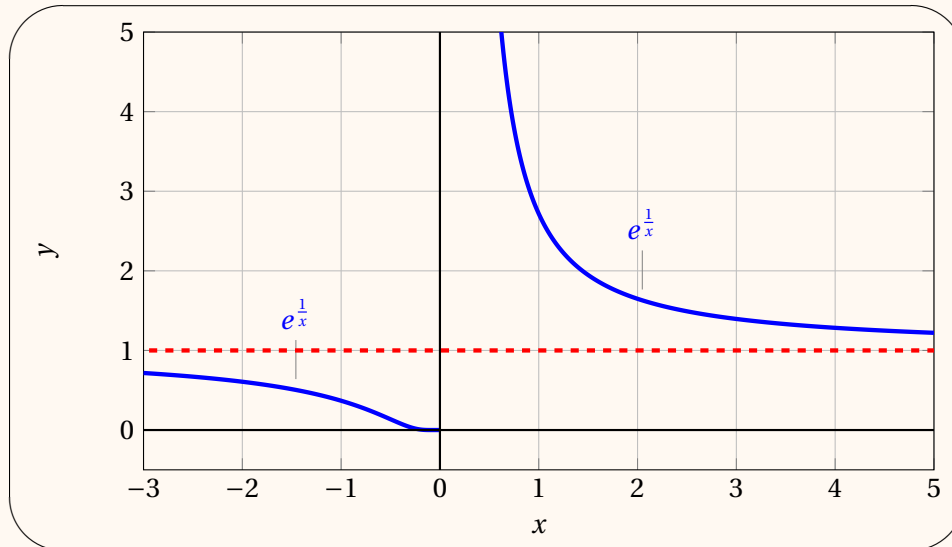
$$y = 2x + 1 \text{ as } x \rightarrow -\infty$$

Example 11: Graphs

$$f(x) = e^x, \quad \lim_{x \rightarrow +\infty} e^x = +\infty, \quad \lim_{x \rightarrow -\infty} e^x = 0, \quad \lim_{x \rightarrow 0} e^x = 1$$

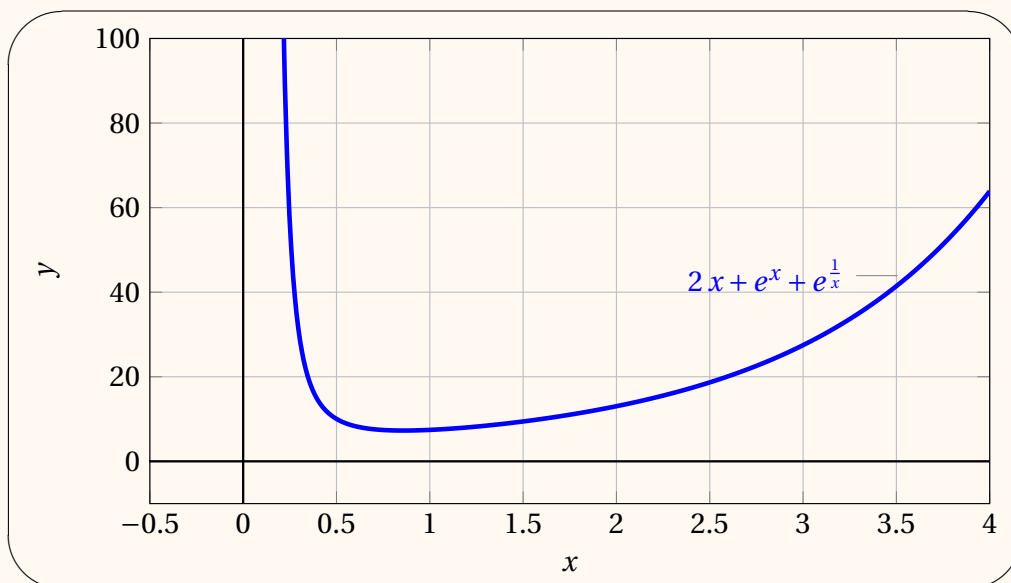


$$f(x) = e^{\frac{1}{x}}, \quad \lim_{x \rightarrow +\infty} e^{\frac{1}{x}} = 1, \quad \lim_{x \rightarrow -\infty} e^{\frac{1}{x}} = 1, \quad \lim_{x \rightarrow 0^-} e^{\frac{1}{x}} = 0, \quad \lim_{x \rightarrow 0^+} e^{\frac{1}{x}} = +\infty$$



Example 12: Actual graph

$$f(x) = 2x + e^x + e^{\frac{1}{x}}, \quad f(x) \sim e^x \text{ as } x \rightarrow +\infty, \quad \lim_{x \rightarrow 0^+} f(x) = +\infty$$



$$f(x) = 2x + e^x + e^{\frac{1}{x}}, \quad f(x) \sim 2x \text{ as } x \rightarrow -\infty, \quad \lim_{x \rightarrow 0^-} f(x) = 1$$

